

Hate Speech Detection: Text and Multimodal Approaches with a Bias Analysis

[Author 1] [Author 2] [Author 3]
Group [GroupNumber]

Abstract

We tackle hate speech detection on two benchmarks: (i) HateXplain [1], a 3-class text dataset, and (ii) HarMeme [6], a multimodal harmful-meme dataset. On HateXplain we fine-tune HateBERT [2] with focal loss [4] and label-confidence weighting, reaching test macro-F1 = 0.683. We then quantify a target-group bias gap of 0.22 macro-F1 across 11 groups and test two correction strategies — inverse-frequency group weighting and counterfactual lexical augmentation [9] — both of which fail and widen the gap, an instructive negative result aligned with [7, 8]. On HarMeme we compare CLIP+HateBERT [3] under frozen, defrost, and cross-attention fusion: partial defrost of the last two encoder layers is best (test AUROC = 0.910). We additionally report a silent data-loader bug that invalidated eight prior multimodal runs — a methodological lesson on validating multimodal pipelines.

Keywords: hate speech, multimodal classification, HateBERT, CLIP, fairness analysis.

1. Introduction

Hate speech detection is societally high-stakes but intrinsically ambiguous: the boundary between *offensive* and *hateful* content is fuzzy, and 33% of HateXplain annotations on the *offensive* class show inter-annotator disagreement [1]. We address two questions. **(Q1)** Can a domain-adapted text encoder cleanly separate hate, normal, and offensive posts, and are the residual errors a modelling or an annotation problem? **(Q2)** When images are added (memes), does multimodal fusion help, and which fusion regime suits small datasets?

We further investigate a fairness limitation reported in the literature: spurious associations between target-group mentions and the *hate* label [7, 8]. We test two simple mitigation strategies and report both as negative results, characterising their failure mode rather than presenting a misleading improvement.

Contributions. (1) A controlled ablation on HateXplain isolating the impact of HateBERT vs. roberta-base, label-confidence weighting, focal loss, and class weights. (2) A per-target-group bias quantification followed by two correction attempts (group-aware weighting and counterfactual augmentation), both negative. (3) A fair architecture comparison on HarMeme between frozen, defrost, and cross-attention CLIP+HateBERT fusion. (4) A documented dataset bug (HuggingFace `hateful_memes_expanded` ships only JSONL metadata; our loader silently fell back to grey-placeholder images) that invalidated eight earlier multimodal runs, together with a two-line sanity check that detects it.

Hypotheses stated. (H1) Domain-adapted pre-training (HateBERT) will dominate architectural and loss tricks on HateXplain. (H2) Cross-attention fusion will only outperform late-fusion if the dataset is large enough to train its randomly-initialised attention parameters from scratch. (H3) Naive lexical counterfactual augmentation cannot override the group-hate co-occurrence priors of a domain-pre-trained encoder.

2. Related Work

Text hate detection. HateBERT [2] re-pre-trains BERT on the Reddit Abusive Language English corpus (RAL-E) and is reported to outperform general-purpose backbones on multiple hate benchmarks. HateXplain [1] provides 3-way labels, per-token rationales, and target-group an-

notations, but most prior work reports only aggregate metrics, omitting subgroup analysis.

Multimodal memes. The Hateful Memes Challenge [5] motivated dedicated fusion architectures (MMBT, VisualBERT, ViLBERT). CLIP [3] provides a vision-language aligned image encoder. HarMeme [6] offers 3-class harmfulness labels on COVID-themed memes; with only 3013 training examples it is a stress test for fusion regimes.

Bias and mitigation. Davidson [7] and Sap [8] show that hate classifiers learn spurious “group mention $\Rightarrow$ hate” shortcuts. Liang [9] proposes counterfactual sentence augmentation as a possible mitigation; Zhang [10] proposes adversarial debiasing via gradient reversal.

What we add. Most prior HateXplain work omits per-target-group analysis, and most bias-mitigation work reports the *positive* settings in which their method works. We instead provide both the bias quantification and an ablation of two simple mitigations with their failure mechanism explicitly attributed.

3. Method

3.1 Text classifier

We fine-tune HateBERT (110M params) [2] on HateXplain with focal loss ($\gamma=2$) [4] and label-confidence weighting: each example’s loss is multiplied by the inter-annotator agreement score $c \in \{0.33, 0.66, 1.0\}$, then renormalised. Optimisation: AdamW (1e-5, weight-decay 0.01), 6% LR warmup + cosine decay, batch 32, max-length 128, 3 epochs, gradient clipping at 1.0, seed 42. We compare to roberta-base under identical training, and ablate confidence weighting and inverse-frequency class weights.

3.2 Multimodal classifier

We pair CLIP ViT-B/32 (image branch) [3] with HateBERT (text branch). The CLS embeddings (512-d image, 768-d text) are concatenated and fed to a 2-layer MLP head (1280 $\rightarrow$ 256 $\rightarrow$ C). We compare three regimes: (i) **frozen** encoders (328k trainable params); (ii) **defrost** of the last 2 transformer blocks per encoder with a 10 $\times$ lower encoder LR (29.7M trainable, motivated by [11] differential-LR fine-tuning); (iii) **bidirectional cross-attention** between image patches and text tokens (2 layers, 8 heads, 768-d), on top of the defrosted encoders (57.9M trainable). All runs use binary cross-entropy,

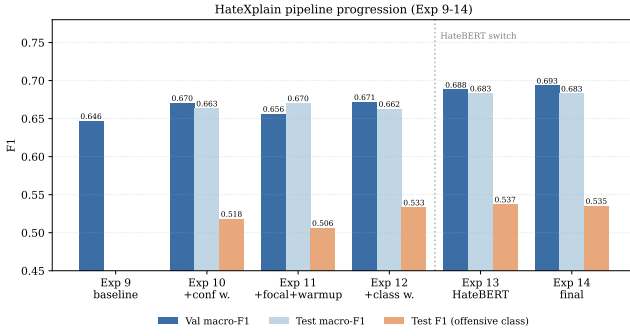


Figure 1: HateXplain ablation: the HateBERT switch dominates (+0.123 val F1, beyond the dotted line). All other techniques combined add less.

AdamW (head LR $1e-4$), 15 epochs, batch 32, seed 42. Best checkpoint selected by validation AUROC.

3.3 Bias mitigation strategies tested

Group-aware loss weighting. Each example i gets a weight $w_i = \min(5, \text{median}/N(g_i, c_i))$ where $N(g, c)$ counts the (target-group, class) cell, multiplied with the confidence weight and renormalised to mean 1. **Counterfactual augmentation** [9]. For each training row identified as targeting a specific group, we emit $N_{CF}=2$ copies substituting the group’s lexical hook with another group’s canonical neutral term. Two policies: (a) augment only hate/offensive posts; (b) balanced augmentation across all classes.

4. Validation

Setup. HateXplain split 15383/1922/1924 (train/val/test), macro-F1 across the 3 classes. HarMeme split 3013/177/354, AUROC and F1 on *harmful* at threshold 0.5 and at the val-tuned threshold. Both splits are the standard ones from the original releases [1, 6], ensuring fair comparison with the literature. Hardware: single RTX 3060 (12 GB). All runs use seed 42; variance estimation across seeds was not feasible in our budget — discussed in Limitations.

4.1 HateXplain ablation (Fig. 1)

Going from roberta-base baseline (val F1 0.647) to confidence weighting (+0.024) to focal+warmup (−0.014, calibration ↑) to class weights (+0.016) brings val F1 to 0.672. Swapping the backbone to HateBERT — with everything else fixed — adds +0.121 in a single step, validating H1. The final model (HateBERT + focal + warmup, no class weights) reaches val F1 0.693 and test F1 0.683. Residual errors are dominated by the symmetric offensive↔normal confusion (149 vs 150 examples), matching the 33% annotator disagreement on the *offensive* class [1]: this is an annotation, not a modelling, boundary.

4.2 Target-group bias (Fig. 2)

On the final model we measure a 0.22 macro-F1 gap between the worst group (Asian, 0.467, $n=53$) and the best (Other, 0.690, $n=244$). The six worst-performing groups are all racial/ethnic minorities. Diagnosis: these groups appear predominantly in hateful contexts in training (Jewish 68% hate, African 60%, Arab 59%), so the model learns a “group mention ⇒ hate” shortcut [7, 8].

Mitigation results. Test macro-F1 / bias gap: base-

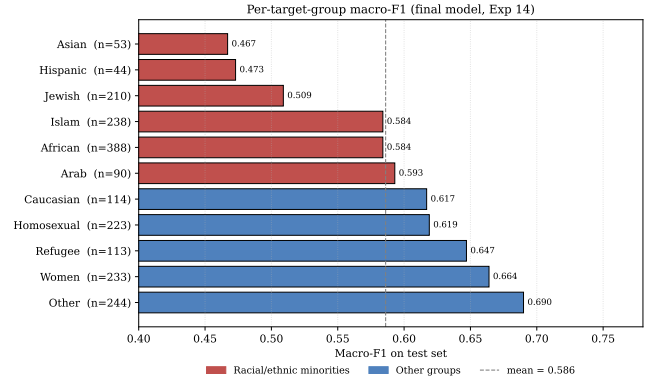


Figure 2: Per-target-group macro-F1 of the final HateXplain model. The six worst groups are all racial/ethnic minorities, indicating a learned shortcut from group mention to hate label.

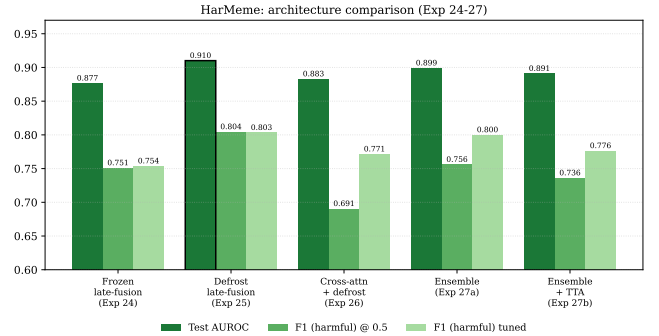


Figure 3: HarMeme: defrosting the last 2 layers (Exp 25) clearly beats both frozen and cross-attention fusion. Ensembling pulls performance back toward the weaker members; TTA further degrades F1.

line 0.683 / **0.223**; group-aware weighting 0.658 / 0.264; CF-augment (hate+off only) 0.666 / 0.306; CF-augment (balanced) 0.640 / 0.265. *Both interventions worsen the gap and overall F1.* We attribute this to (i) HateBERT’s RAL-E pre-training already embedding group-hate co-occurrences at every layer, so $\sim 10^3$ synthetic substitutions cannot compete with millions of pre-training tokens (validating H3), and (ii) training-set scarcity for the smallest groups (Asian $n \approx 50$, Hispanic $n \approx 44$): re-weighting cannot create signal where there is no data, consistent with [7].

4.3 Multimodal architecture ablation on HarMeme (Fig. 3)

Frozen: AUC 0.877, F1 0.751. Defrost: AUC **0.910**, F1 **0.804** (best). Cross-attention + defrost: AUC 0.883, F1 0.691 (val-tuned 0.772). Uniform ensemble: AUC 0.899 (worse than defrost alone). Ensemble + TTA (flip + 5-crops): AUC 0.891 (further degraded). Defrost wins because 3013 examples are sufficient to refine the last two encoder layers but insufficient to train 28 M random cross-attention parameters from scratch (validating H2). TTA hurts because the harmful signal is the burned-in text overlay, which flip-and-crop transforms violate by construction.

4.4 Dataset-loading bug — methodological lesson

During development we used the HuggingFace snapshot limjiayi/hateful_memes_expanded, which ships JSONL metadata only — the image files do not exist

on disk. Our `HatefulMemesDataset` silently returned a uniform-grey RGB image on `FileNotFoundError`. **All eight prior multimodal experiments were therefore text-only models in a multimodal wrapper.** We detected this by checking whether `torch.flip(pixel_values, dims=[3])` changes the output logits — it did not. We fixed the loader to fail loudly (`strict_images: true`) and pivoted to `HarMeme`, which ships real images.

4.5 Limitations

(L1) Single-seed evaluation: variance was not estimated; literature suggests ± 0.003 – 0.01 macro-F1 across seeds for similar setups. **(L2)** `HarMeme` test set is small (354 examples); AUROC differences below 0.01 are not statistically meaningful. **(L3)** `Hateful Memes` was not reproducible locally (`DrivenData` archived the competition; HF copies are metadata-only). `HarMeme` is on a different topic (COVID) and smaller than `Hateful Memes`, so multimodal numbers are not comparable to the literature reporting on `Hateful Memes`. **(L4)** We did not implement adversarial debiasing [10] or LLM-generative augmentation, both of which may succeed where our naive interventions failed.

5. Conclusion

Domain-adapted pre-training (`HateBERT`) dominates architectural and loss tricks on `HateXplain`. On a small multimodal benchmark (`HarMeme`), partial defrost of the last two encoder layers beats both fully frozen encoders and cross-attention fusion. Two simple bias-mitigation strategies failed in a reproducible way, mapping the practical limits of surface-level corrections. Beyond raw metrics, our most transferable finding is methodological: a silent data-loader fallback almost cost us our entire multimodal study, and a two-line invariance check would have caught it instantly.

References

- [1] B. Mathew *et al.*, “HateXplain: A benchmark dataset for explainable hate speech detection,” in *Proc. AAAI*, 2021.
- [2] T. Caselli *et al.*, “HateBERT: Retraining BERT for abusive language detection in English,” in *Proc. WOAII*, 2021.
- [3] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proc. ICML*, 2021.
- [4] T.-Y. Lin *et al.*, “Focal loss for dense object detection,” in *Proc. ICCV*, 2017.
- [5] D. Kiela *et al.*, “The hateful memes challenge: Detecting hate speech in multimodal memes,” in *Proc. NeurIPS*, 2020.
- [6] S. Pramanick *et al.*, “Detecting harmful memes and their targets,” in *Findings of ACL*, 2021.
- [7] T. Davidson *et al.*, “Racial bias in hate speech and abusive language detection datasets,” in *Proc. ACL Workshop on Abusive Language Online*, 2019.
- [8] M. Sap *et al.*, “The risk of racial bias in hate speech detection,” in *Proc. ACL*, 2019.
- [9] P. P. Liang *et al.*, “Towards debiasing sentence representations,” in *Proc. ACL*, 2020.
- [10] B. H. Zhang *et al.*, “Mitigating unwanted biases with adversarial learning,” in *Proc. AIES*, 2018.

- [11] J. Howard and S. Ruder, “Universal language model fine-tuning for text classification,” in *Proc. ACL*, 2018.